Foundations of Precision and recall

[Precision and recall]

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}$$

1.0 Content Block

Multi-class evaluation Precision and recall values can also be calculated for classification problems with more than two classes. To obtain the precision for a given class, we divide the number of true positives by the classifier bias towards this class (number of times that the classifier has predicted the class). To calculate the recall for a given class, we divide the number of true positives by the prevalence of this class (number of times that the class occurs in the data sample). The class-wise precision and recall values can then be combined into an overall multi-class evaluation score, e.g., using the macro F1 metric.

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In pattern recognition, information retrieval, object detection and classification (machine learning), precision and recall are performance metrics that apply to data retrieved from a collection, corpus or sample space. Precision (also called positive predictive value) is the fraction of relevant instances among the retrieved instances. Written as a formula:

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Definition For classification tasks, the terms true positives, true negatives, false positives, and false negatives compare the results of the classifier under test with trusted external judgments. The terms positive and negative refer to the classifier's prediction (sometimes known as the expectation), and the terms true and false refer to whether that prediction corresponds to the external judgment (sometimes known as the observation). Let us define an experiment from P positive instances and N negative instances for some condition. The four outcomes can be formulated in a 2×2 contingency table or confusion matrix, as follows:

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Two other commonly used F {\displaystyle F} measures are the F 2 {\displaystyle F_{2}} measure, which weights recall higher than precision, and the F 0.5 {\displaystyle F_{0.5}} measure, which puts more emphasis on precision than recall. The F-measure was derived by van Rijsbergen (1979) so that F β {\displaystyle F_{\beta }} "measures the effectiveness of retrieval with respect to a user who attaches β {\displaystyle \beta } times as much importance to recall as precision". It is based on van Rijsbergen's effectiveness measure E α = 1 − 1 α P + 1 − α R {\displaystyle E_{\alpha }=1-{\frac {1}{{\frac {\alpha }{P}}+{\frac {1-\alpha }{R}}}}} , the second term being the weighted harmonic mean of precision and recall with weights (α , 1 − α) {\displaystyle (\alpha ,1-\alpha)} . Their relationship is F β = 1 − E α {\displaystyle F_{\beta }=1-E_{\alpha }} where α = 1 1 + β 2 {\displaystyle \alpha ={\frac {1}{1+\beta ^{2}}}} .

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Quantitative finance. A model may output a continuous buy/sell signal proportional to its conviction for each asset at a given moment. Applying a fixed threshold (e.g. signal > 0.1 = buy) discards conviction magnitude and conflates strong and mild signals, limiting the model's ability to correctly distinguish a strong buy from a mild one. DNA methylation analysis. Sequencing tools detect differentially methylated regions whose underlying signal is continuous, typically ranging from −1 to +1. Forcing this into a binary framework introduces similar distortions by treating all positive deviations as equivalent regardless of magnitude.

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F β = (1 + β 2) · p r e c i s i o n · r e c a l l β 2 · p r e c i s i o n + r e c a l l {\displaystyle F_{\beta }=(1+\beta ^{2})\cdot {\frac {\mathrm {precision} \cdot \mathrm {recall} }{\beta ^{2}\cdot \mathrm {precision} +\mathrm {recall} }}}

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Precision vs. recall Both precision and recall may be useful in cases where there is imbalanced data. However, it may be valuable to prioritize one metric over the other in cases where the outcome of a false positive or false negative is costly. For example, in medical diagnosis, a false positive test can lead to unnecessary treatment and expenses. In this situation, it is useful to value precision over recall. In other cases, the cost of a false negative is high, and recall may be a more valuable metric. For instance, the cost of a false negative in fraud detection is high, as failing to detect a fraudulent transaction can result in significant financial loss.

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Generalisation to continuous distributions The classical definitions of precision, recall, and F-score are formulated for binary classification, where each instance is either true or false. Many practical domains, however, produce inherently continuous signals rather than discrete labels. Generalising these metrics to continuous distributions enables their direct application to such data without the information loss introduced by thresholding. Two motivating examples illustrate the limitation of binary frameworks:

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Thus the precision has an explicit dependence on r {\textstyle r} . Starting with balanced classes at r = 1 {\textstyle r=1} and gradually decreasing r {\textstyle r} , the corresponding precision will decrease, because the denominator increases. Another metric is the predicted positive condition rate (PPCR), which identifies the percentage of the total population that is flagged. For example, for a search engine that returns 30 results (retrieved documents) out of 1,000,000 documents, the PPCR is 0.003%.

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Probabilistic definition Precision and recall can be interpreted as (estimated) conditional probabilities: Precision is given by P (C = P | C ^ = P) {\displaystyle \mathbb {P} (C=P|{\hat {C}}=P)} while recall is given by P (C ^ = P | C = P) {\displaystyle \mathbb {P} ({\hat {C}}=P|C=P)} , where C ^ {\displaystyle {\hat {C}}} is the predicted class and C {\displaystyle C} is the actual class (i.e. C = P {\displaystyle C=P} means the actual class is positive). Both quantities are, therefore, connected by Bayes' theorem.

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In a classification task, the precision for a class is the number of true positives (i.e. the number of items correctly labelled as belonging to the positive class) divided by the total number of elements labelled as belonging to the positive class (i.e. the sum of true positives and false positives, which are items incorrectly labelled as belonging to the class). Recall in this context is defined as the number of true positives divided by the total number of elements that actually belong to the positive class (i.e. the sum of true positives and false negatives, which are items which were not labelled as belonging to the positive class but should have been). Precision and recall are not particularly useful metrics when used in isolation. For instance, it is possible to have perfect recall by simply retrieving every single item. Likewise, it is possible to achieve perfect precision by selecting only a very small number of extremely likely items. In a classification task, a precision score of 1.0 for a class C means that every item labelled as belonging to class C does indeed belong to class C (but says nothing about the number of items from class C that were not labelled correctly) whereas a recall of 1.0 means that every item from class C was labelled as belonging to class C (but says nothing about how many items from other classes were incorrectly also labelled as belonging to class C). Often, there is an inverse relationship between precision and recall, where it is possible to increase one at the cost of reducing the other, but context may dictate if one is more valued in a given situation: A smoke detector is generally designed to commit many Type I errors (to alert in many situations when there is no danger), because the cost of a Type II error (failing to sound an alarm during a major fire) is prohibitively high. As such, smoke detectors are designed with recall in mind (to catch all real danger), even while giving little weight to the losses in precision (and making many false alarms). In the other direction, Blackstone's ratio, "It is better that ten guilty persons escape than that one innocent suffer," emphasizes the costs of a Type I error (convicting an innocent person). As such, the criminal justice system is geared toward precision (not convicting innocents), even at the cost of losses in recall (letting more guilty people go free). A brain surgeon removing a cancerous tumor from a patient's brain illustrates the tradeoffs as well: The surgeon needs to remove all of the tumor cells since any remaining cancer cells will regenerate the tumor. Conversely, the surgeon must not remove healthy brain cells since that would leave the patient with impaired brain function. The surgeon may be more liberal in the area of the brain they remove to ensure they have extracted all the cancer cells. This decision increases recall but reduces precision. On the other hand, the surgeon may be more conservative in the brain cells they remove to ensure they extract only cancer cells. This decision increases precision but reduces recall. That is to say, greater recall increases the chances of removing healthy cells (negative outcome) and increases the chances of removing all cancer cells (positive outcome). Greater precision decreases the chances of removing healthy cells (positive outcome) but also decreases the chances of removing all cancer cells (negative outcome). Usually, precision and recall scores are not discussed in isolation. A precision-recall curve plots precision as a function of recall; usually precision will decrease as the recall increases. Alternatively, values for one measure can be compared for a fixed level at the other measure (e.g. precision at a recall level of 0.75) or both are combined into a single measure. Examples of measures that are a combination of precision and recall are the F-measure (the weighted harmonic mean of precision and recall), or the Matthews correlation coefficient, which is a geometric mean of the chance-corrected variants: the regression coefficients Informedness (DeltaP') and Markedness (DeltaP). Accuracy is a weighted arithmetic mean of Precision and Inverse Precision (weighted by Bias) as well as a weighted arithmetic mean of Recall and Inverse Recall (weighted by Prevalence). Inverse Precision and Inverse Recall are simply the Precision and Recall of the inverse problem where positive and negative labels are exchanged (for both real classes and prediction labels). True Positive Rate and False Positive Rate, or equivalently Recall and 1 - Inverse Recall, are frequently plotted against each other as ROC curves and provide a principled mechanism to explore operating point tradeoffs. Outside of Information Retrieval, the application of Recall, Precision and F-measure are argued to be flawed as they ignore the true negative cell of the contingency table, and they are easily manipulated by biasing the predictions. The first problem is 'solved' by using Accuracy and the second problem is 'solved' by discounting the chance component and renormalizing to Cohen's kappa, but this no longer affords the opportunity to explore tradeoffs graphically. However, Informedness and Markedness are Kappa-like renormalizations of Recall and Precision, and their geometric mean Matthews correlation coefficient thus acts like a debiased F-measure.